

TECHNICAL REPORT

Weakly Supervised Semantic Segmentation for Remote Sensing Images Using Partial Cross Entropy Loss

Point-Supervised UNet Segmentation on the LandCover.AI Dataset

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Abstract

Semantic segmentation is a fundamental task in remote sensing image analysis, where each pixel in an image is assigned a semantic category. Traditional segmentation methods require dense pixel-level annotations, which are expensive and time-consuming to obtain. This project investigates a weakly supervised segmentation approach based on point annotations rather than complete segmentation masks.

A UNet architecture was employed as the segmentation backbone and trained using a Partial Cross Entropy (PFCE) loss function that utilizes only labeled pixels while ignoring unlabeled regions. The LandCover.AI dataset was used for experimentation, and point annotations were simulated from the original segmentation masks. Multiple experiments were conducted to evaluate the impact of annotation density on model performance.

The results demonstrate the feasibility of learning segmentation models from sparse annotations while significantly reducing annotation requirements.

1. Introduction

Semantic segmentation aims to classify every pixel in an image into a predefined semantic category. It plays a critical role in numerous applications such as satellite image analysis, environmental monitoring, urban planning, autonomous driving, and medical image analysis.

Despite recent advances in deep learning-based segmentation, most approaches rely on fully annotated segmentation masks. Producing these masks requires extensive manual labeling efforts, making large-scale segmentation projects costly and difficult to maintain.

To address this challenge, weakly supervised learning approaches have emerged. Instead of requiring full pixel-level annotations, weak supervision utilizes sparse annotations such as image labels, bounding boxes, scribbles, or point annotations.

In this project, point-based supervision is investigated for remote sensing image segmentation. A Partial Cross Entropy loss function is employed to train a segmentation network using only a small subset of labeled pixels while ignoring the remaining unlabeled regions.

2. Dataset Description

The experiments were conducted using the LandCover.AI dataset, which contains aerial remote sensing images together with corresponding land cover segmentation masks.

The dataset consists of high-resolution RGB images and pixel-level annotations representing different land cover classes. The original masks contain dense supervision; however, to simulate a realistic weakly supervised scenario, only a limited number of pixels were randomly selected from each class and used as training labels.

Prior to training, all images and masks were resized to 512×512 pixels to ensure consistent input dimensions and efficient batch processing.

Property	Value
Dataset	LandCover.AI
Image Type	RGB
Input Size	512×512
Number of Classes	5
Annotation Type	Simulated Point Labels
Framework	PyTorch

Table 1: Summary of dataset properties used in the experiments.

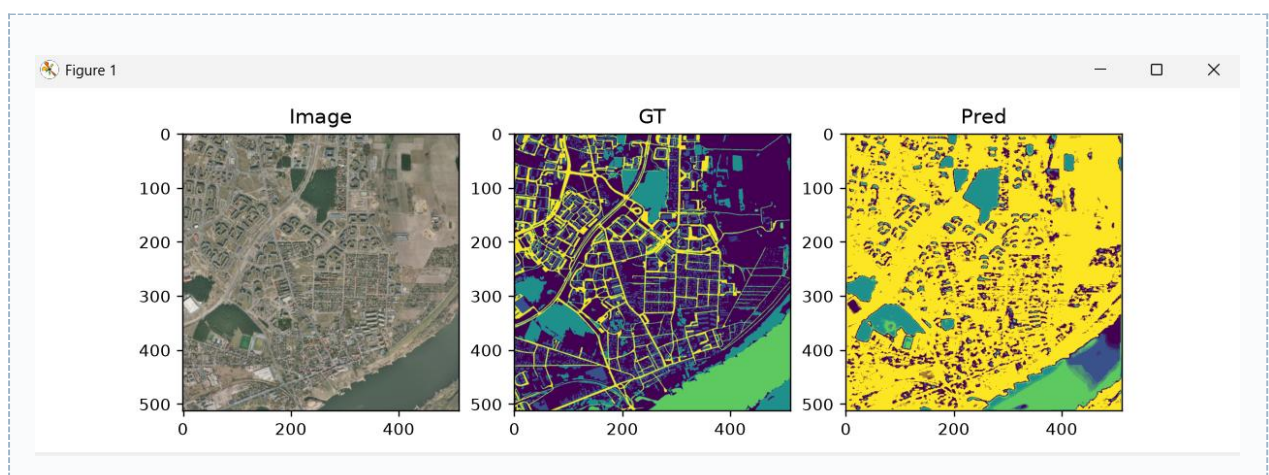


Figure 1: Example image and corresponding segmentation mask from the LandCover.AI dataset.

3. System Architecture

The proposed framework consists of four major stages:

- Dataset Preparation
- Point Annotation Simulation
- Segmentation Network
- PFCE-Based Training

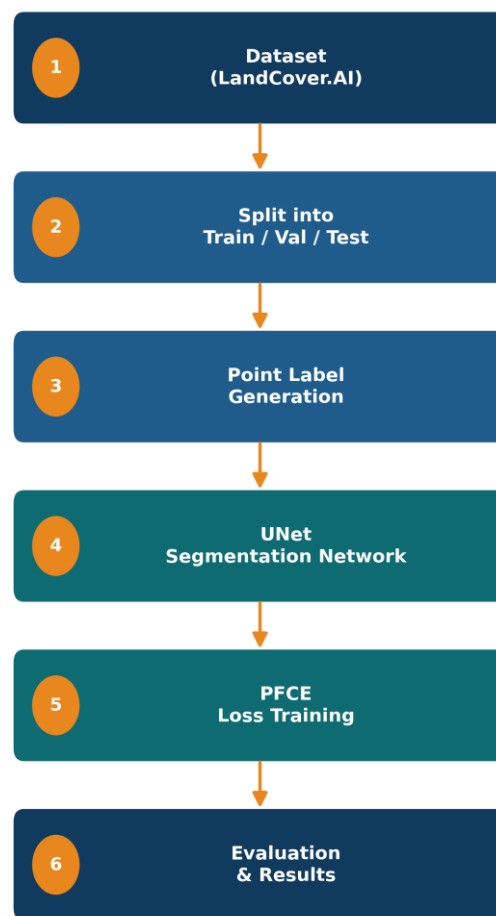


Figure 2: End-to-end processing pipeline, from raw dataset to final evaluation.

3.1 Dataset Preparation

A custom dataset loader was implemented using PyTorch's Dataset class. The loader is responsible for reading image-mask pairs, resizing images and masks, converting images into tensors, and preparing training samples.

The dataset was divided into training, validation, and testing subsets using a custom splitting script.

3.2 Point Annotation Simulation

Since the objective is weakly supervised segmentation, complete segmentation masks were not directly used during training.

Instead, a point annotation generation procedure was implemented. For each semantic class present in a mask, a fixed number of pixels were randomly sampled and retained as labeled points. All remaining pixels were assigned a value of -1 to indicate unlabeled regions.

This process simulates realistic sparse annotations while maintaining a controllable supervision density.

3.3 Segmentation Model

The segmentation model employed in this project is UNet.

UNet is a convolutional encoder-decoder architecture specifically designed for dense prediction tasks. The encoder extracts hierarchical features from the input image, while the decoder progressively reconstructs spatial information to generate pixel-level predictions.

The model consists of:

- Encoder blocks
- Max-pooling layers
- Bottleneck layer
- Decoder blocks
- Skip connections

The skip connections help preserve fine spatial details and improve segmentation accuracy.

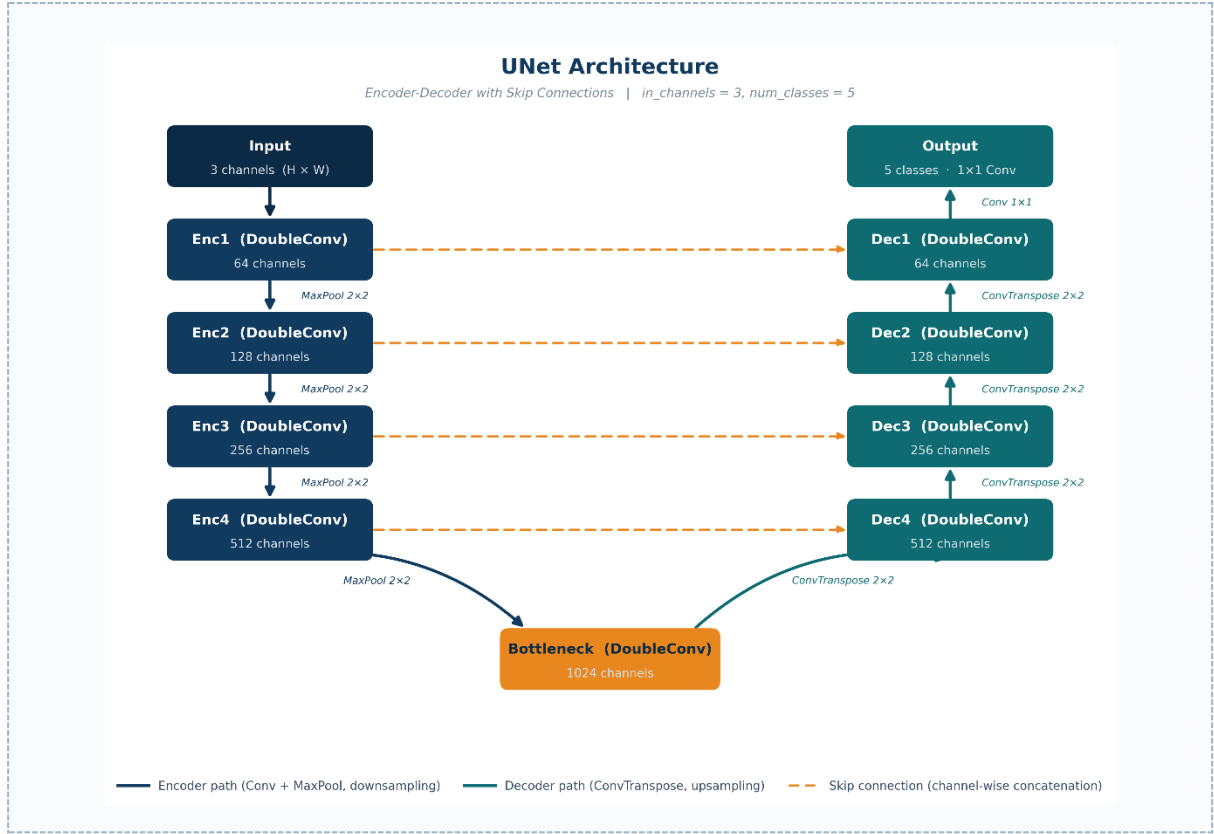


Figure 5: UNet architecture used as the segmentation backbone.

4. Partial Cross Entropy Loss

Traditional Cross Entropy computes the loss over all image pixels. However, in weakly supervised learning, labels are only available for a subset of pixels.

To address this limitation, Partial Cross Entropy (PFCE) was implemented. The loss is computed only over labeled pixels while ignoring unlabeled regions, as defined below:

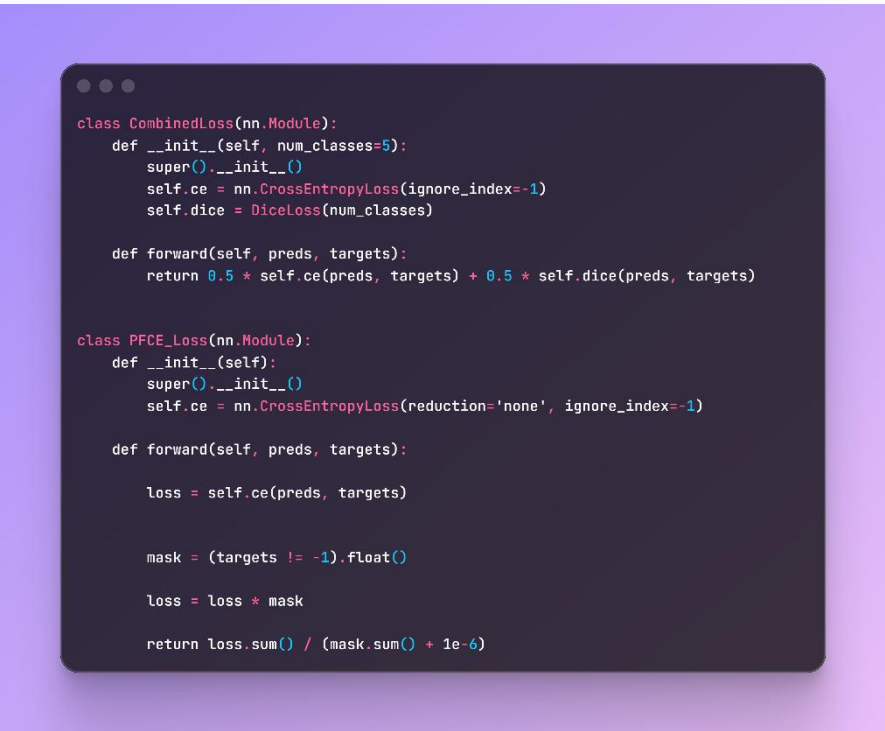
Equation 1

$$\mathcal{L}_{PFCE} = \frac{\sum (FocalLoss(Pred, GT) \times Mask_{labeled})}{\sum Mask_{labeled}}$$

$Pred$ = model prediction GT = ground truth label $Mask_{labeled}$ = labeled-pixel mask

Equation 1: Partial Cross Entropy (PFCE) loss, computed only over labeled pixels.

This formulation allows the network to learn from sparse supervision while avoiding incorrect penalties on unlabeled pixels.



```
class CombinedLoss(nn.Module):
    def __init__(self, num_classes=5):
        super().__init__()
        self.ce = nn.CrossEntropyLoss(ignore_index=-1)
        self.dice = DiceLoss(num_classes)

    def forward(self, preds, targets):
        return 0.5 * self.ce(preds, targets) + 0.5 * self.dice(preds, targets)

class PFCE_Loss(nn.Module):
    def __init__(self):
        super().__init__()
        self.ce = nn.CrossEntropyLoss(reduction='none', ignore_index=-1)

    def forward(self, preds, targets):

        loss = self.ce(preds, targets)

        mask = (targets != -1).float()

        loss = loss * mask

        return loss.sum() / (mask.sum() + 1e-6)
```

Figure 6: Implementation of the PFCE loss logic (*losses.py*).

5. Experimental Setup

All experiments were conducted using the same model architecture and training settings. The following configuration was used:

Parameter	Value
Model	UNet
Loss	PFCE
Optimizer	Adam
Learning Rate	1e-4
Epochs	5
Batch Size	2

Parameter	Value
Image Size	512 × 512

Table 2: Training configuration used across all experiments.

6. Experiments

The objective of the experiments was to investigate how the number of labeled points affects segmentation performance.

Hypothesis

Increasing the number of labeled points should provide richer supervision and improve segmentation performance.

Experiment 1: Effect of Point Density

Three configurations were evaluated, with all other parameters held unchanged:

- 10 points per class
- 50 points per class
- 100 points per class

Number of Points	Validation Loss
10	1.5953
50	1.5983
100	1.5932

Table 3: Validation loss for different point annotation densities.

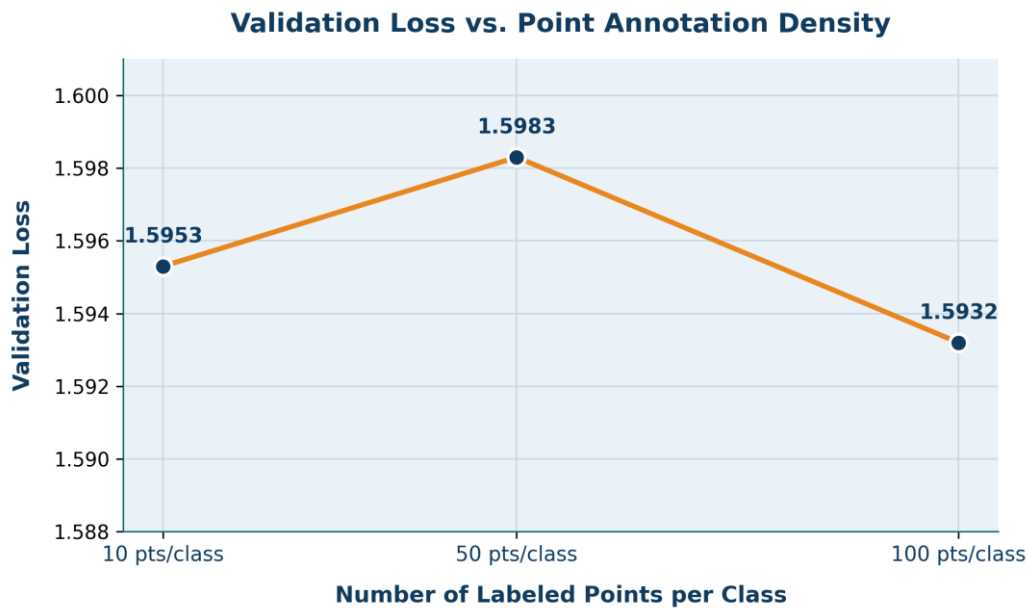


Figure 7: Validation loss across the three point-density configurations (10, 50, and 100 points per class).

7. Results and Discussion

The experimental results indicate that the model successfully learns from sparse point annotations. However, the improvement achieved by increasing the number of points was relatively small.

Several factors may explain this observation:

- The training duration was limited.
- Point locations were selected randomly.
- Sparse annotations provide limited spatial information.
- Additional regularization techniques were not employed.

Although performance gains were modest, the results demonstrate that meaningful segmentation learning can be achieved using only a small fraction of the original annotations.

8. Project Structure and Implementation

The project was organized into modular components to improve maintainability and reproducibility:

File	Purpose
dataset.py	Dataset loading and preprocessing

File	Purpose
model.py	UNet implementation
losses.py	PFCE and Dice losses
train.py	Training loop
metrics.py	Evaluation metrics
experiments.py	Experimental analysis
visualize.py	Visualization of predictions
split.py	Dataset splitting

Table 4: Modular project file structure.

This modular design enables easy experimentation and future extensions.

9. Conclusion

This work presented a weakly supervised semantic segmentation framework for remote sensing imagery using point annotations. A UNet architecture combined with a Partial Cross Entropy loss was successfully implemented and evaluated on the LandCover.AI dataset.

Experimental results demonstrated that segmentation models can be trained using sparse annotations while significantly reducing annotation costs. Future work may explore semi-supervised learning, pseudo-label generation, stronger backbone networks, and advanced augmentation strategies to further improve performance.

10. Project Repository

This project's implementation is maintained in a version-controlled repository, allowing reproducibility and future collaboration.

PROJECT REPOSITORY

The full source code, notebooks, and trained model checkpoints for this project are available on GitHub:

[\[Insert your GitHub repository link here, e.g. `https://github.com/your-username/your-repo` \]](#)

11. References

1. LandCover.AI Dataset. <https://huggingface.co/datasets/dragon7/LandCover.ai>